

Storage Units · AS · Answers

24 written + 6 pen-first = 30 · Student copy

One mark per stated point unless grouped. Accept equivalent correct working; do not credit contradictory working. NE = not enough. Uncompressed payload assumptions follow the question sheet.

1. Units [4]

(a) Byte **8 bits** [1]; nibble **4 bits** [1]. (b) **8000 bits** [1]. (c) **8192 bits** [1].

2. Bitmap [4]

(a) $1024 \times 768 \times 24 = 18\,874\,368$ bits [1]; $\div 8 = 2\,359\,296$ bytes [1]; $\div 1024^2 = \mathbf{2.25\,MiB}$ [1]. Equivalent correct methods accepted, max two working plus one answer. (b) Any one: width, height/dimensions, colour depth, format [1]. The pixel-data estimate excludes the header.

3. Sound [4]

(a) $16\,384 \times 16 \times 32 = 8\,388\,608$ bits [1]; $\div 8 = 1\,048\,576$ bytes [1]; $\div 1024 = \mathbf{1024\,KiB}$ [1]. Equivalent valid byte-per-sample/second stages accepted. (b) **512 KiB** [1].

4. Unit error [3]

Missing bits-to-bytes conversion/division by 8 [1]; $3\,840\,000 \div 8 = 480\,000$ bytes [1]; $\div 1024 = \mathbf{468.75\,KiB}$ [1].

5. Prefixes [1]

1 GB = 1 000 000 000 bytes; **1 GiB = 1 073 741 824 bytes** (10^9 and 2^{30} accepted). Both required [1].

6. Authored prefix rehearsal [4]

(a) **binary** [1]; $5 \times 1024^2 = \mathbf{5\,242\,880\,bytes}$ [1]. (b) **decimal/denary** [1]; $7 \times 1000 = \mathbf{7000\,gigabytes}$ [1]. Decimal terabytes-to-gigabytes uses 1000; binary mebibytes-to-bytes uses 1024^2 .

7. Compare [2]

$3100\,kB = 3\,100\,000$ bytes and $3\,MiB = 3\,145\,728$ bytes, both counts correct [1]; **3 MiB is larger** [1].

8. Whole file [2]

$1024\,bytes = 1\,KiB$ [1]; $512 + 1 = \mathbf{513\,KiB}$ [1]. Applies to the stated no-compression/no-other-overhead model.

9. Pen-first file-size machine [6]

(a) $512 \times 512 \times 16 = 4\,194\,304$ bits; $\div 8 \div 1024$ [1] = **512 KiB** [1]. (b) **1024 KiB** [1]; twice as many pixels, with the same bits per pixel [1]. (c) **256 KiB** [1]; each pixel uses half as many bits, with the same pixel count [1]. All six marks are human marked. Running the cell is a check, not an extra mark.

Extension (unmarked): estimate $\text{width} \times \text{height} \times 24 \div 8$ bytes. Actual depth, compression, headers and metadata may differ from the assumed model. Dimensions alone do not establish the encoded file size. Karel 07/08 remains outside the marks.